

Stanley College of Engineering & Technology for Women (A)**B.E (Common to all) II Semester (Main) Examinations-September-2023****Mathematics-II****Time: 03Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any four out of remaining six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question Paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. Solve $(y^2 - 2xy) dx + (2xy - x^2) dy = 0$ [2M CO1BTL2]
 b. Solve $(D^3 - 3D + 2) y = 0$ [2M CO2 BTL3]
 c. Evaluate $\int_0^1 x^5(1-x)^3 dx$ [2M CO3 BTL4]
 d. Define Rank of Matrix with example. [2M CO4BTL1]
 e. Find $L \{ t^2 \sin 2t \}$ [2M CO5 BTL3]
 f. Find the value of k such that the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2. [2M CO4 BTL2]

Part-B**4X12=48M**

2. a) Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$ [6M CO1 BTL3]
 b) Prove that the system of parabolas $y^2 = 4a(x+a)$ is self-orthogonal. [6M CO1 BTL4]
3. a) Solve $(D^2 + 3D + 2) y = 2 \cos(2x+3) + 2e^x + x^2$ [6M CO2 BTL4]
 b) Solve the differential equation $(D^2 + 4) y = \sec 2x$ by the method of variation of parameters [6M CO2 BTL3]
4. a) State and Prove relation between Beta and Gamma functions [8M CO3 BTL4]
 b) Evaluate $\int_0^\infty x^2 e^{-x^2} dx$ [4M CO3 BTL3]
5. a) Determine whether the following equations have a non-trivial solution, if so solve them: $4x + 2y + z + 3w = 0$, $6x + 3y + 4z + 7w = 0$, $2x + y + w = 0$ [6M CO4 BTL3]
 b) Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$ [6M CO4 BTL4]
6. a) Evaluate $L \left\{ \frac{\cos 2t - \cos 3t}{t} \right\}$ [4M CO5 BTL3]
 b) Evaluate $L^{-1} \left\{ \frac{s^2}{(s^2+4)(s^2+25)} \right\}$ using convolution theorem. [8M CO5 BTL4]
7. a) Solve $P = \sin(y - xp)$. Also find its Singular solution. [4M CO2 BTL2]
 b) Find the nature of the quadratic form $2x^2 + 2y^2 + 2z^2 + 2yz$ [8M CO4 BTL3]



Hall Ticket No. -

Code: R123924

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/IT/CME/AI&DS) II Semester (Main) Examinations-September-2023

Data Structures with C

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- List any four applications of data structures. [2M CO1 BTL1]
 - Write a recursive algorithm for finding factorial of a number [2M CO2 BTL2]
 - The size of the stack is 7. The operations performed are push(35), push(40), push(45), pop(), push(50), pop(), push(55), pop(), pop(). Evaluate the final elements present in the stack. [2M CO7 BTL5]
 - What are the conditions for a balanced binary search tree? [2M CO9 BTL2]
 - Define Hash function. [2M CO9 BTL1]
 - Differentiate a tree and a graph. [2M CO12 BTL2]

Part-B

4X12=48M

- Illustrate asymptotic notations with examples. [6M CO1 BTL3]
 - Write the steps for Binary search algorithm and compare the time complexity with linear search. [6M CO5,12 BTL2]
- Evaluate the following infix expression using Stack:
 $2 * (5 * (3 + 6)) / 5 - 2$ [6M CO7 BTL5]
 - Explain the implementation of queues using arrays. [6M CO8 BTL2]
- What are the applications of linked list data structure? [6M CO8 BTL2]
 - Compare circular linked list and doubly linked list. [6M CO7,8,12 BTL2]
- What are the variants of Binary trees? Explain. [6M CO9 BTL2]
 - Compare BFS and DFS with an example. [6M CO9 BTL2]
- Explain all the operations of m-way search trees. [8M CO10 BTL2]
 - What is linear probing? Explain. [4M CO9 BTL2]
- Sort the following numbers using quick sort:
32,19,20,45,65,12,50,18 [6M CO4 BTL4]
 - Explain the rotations of AVL tree. [6M CO10 BTL2]

Hall Ticket No. -

Code: R123930

Stanley College of Engineering & Technology for Women (A)

B.E(CSE/CME/AI&DS) II Semester (Main) Examinations-September-2023

Environmental Science

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of the remaining Six questions.
ii. Answer to each question must be written in one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Define environment and environmental studies. [2M CO1 BTL1]
b. Explain food web with example. [2M CO2 BTL2]
c. Define Genetic Level Diversity. [2M CO3 BTL1]
d. List the measures to control noise pollution. [2M CO4 BTL1]
e. Explain global warming. [2M CO5 BTL2]
f. What is watershed management? [2M CO5 BTL2]

Part-B

4X12=48M

2. a) Summarize the impact of modern agriculture on the environment. [6M CO1 BTL2]
b) Describe the multidisciplinary nature of environmental studies. [6M CO1 BTL1]
3. a) Define Ecosystem, write the classification of ecosystem. [6M CO2 BTL1]
b) Sketch and explain energy flow in an ecosystem. [6M CO2 BTL2]
4. a) Illustrate Bio-geographical classification of India. [6M CO3 BTL3]
b) Explain endangered and endemic species? List the endangered and endemic species of India. [6M CO3 BTL3]
5. a) Discuss causes, effects, and control measures of Water pollution. [6M CO4 BTL2]
b) Explain salient features of the Water Act, 1974. [6M CO4 BTL4]
6. a) Summarize the different types of disasters. Discuss their ill effects on environment, infrastructure and development. [6M CO5 BTL5]
b) Classify the Disaster management cycle. [6M CO5 BTL4]
7. a) Illustrate the causes, effects, and control measures of Ozone layer depletion. [6M CO5 BTL4]
b) Discuss various renewable and non-renewable energy resources and write a note on Conservation of resources. [6M CO1 BTL2]

Hall Ticket No. :

Code: R123927

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME/AI&DS) II Semester (Main) Examinations-September-2023

Basics Electrical & Electronic Circuits

Time: 03Hours

Max. Marks-60

- Note: i. First Question is Compulsory. Answer any Four out of remaining six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

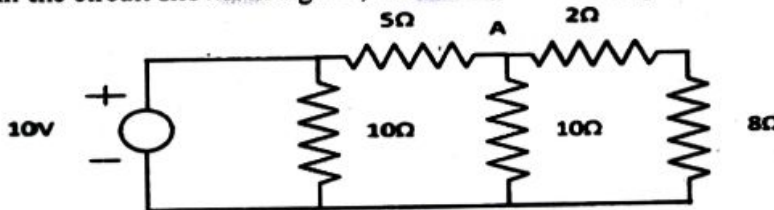
1. a. State Kirchoff's Current law.
b. Define RMS value.
c. List some applications of zener diode.
d. Draw the V-I characteristics of JFET.
e. What are the properties of feedback amplifier?
f. Define real power and reactive power.

CO1 L1 2M
CO2 L1 2M
CO3 L1 2M
CO4 L1 2M
CO5 L1 2M
CO2 L1 2M

Part-B

4X12=48M

2. a. In the circuit shown in figure, find the current through 8Ω branch.



CO1 L3 6M

- b. State Thevenin theorem and explain with an example.
3. a. Derive the expressions for R.M.S and average value of a sinusoidal alternating quantity.
b. Derive the relation between line and phase quantities for star connected balanced systems.
4. a. Draw the circuit diagram for half-wave rectifier and explain its working
b. Sketch and explain in detail about the V - I characteristic of the p - n junction diode.

CO1 L2 6M

CO2 L2 6M

CO2 L2 6M

CO3 L2 6M

CO3 L3 6M

5. a. Explain in detail about Common base circuit configuration with a neat diagram. CO4 L2 6M
- b. With the help of neat sketches and characteristic curves explain the construction & operation of a JFET and mark the regions of operation on the characteristics. CO4 L2 6M
- 6 a. Derive the expression for output resistance of a voltage amplifier circuit. CO5 L3 6M
- b. Draw the circuit diagram of differentiator by using OP-AMP and explain its operation. CO5 L3 6M
- 7 a. With the help of neat circuit diagram, explain the operation of Bridge rectifier. CO5 L2 6M
- b. Sketch and explain in detail about the CRO block diagram. CO3 L3 6M

Hall Ticket No. -

Code: R123922

Stanley College of Engineering & Technology for Women (A)

B.E (CSE/CME/AI&DS) II Semester (Main) Examinations-September-2023

Chemistry

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
 ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. State Reverse Osmosis. [2M CO1 BTL1]
- b. Write about the thermal conversion of solar energy. [2M CO2 BTL1]
- c. How is Galvanic Corrosion explained? [2M CO1 BTL2]
- d. How is Degree of polymerization defined? [2M CO3 BTL2]
- e. Differentiate LCV and HCV. [2M CO4 BTL4]
- f. Give any four applications of Bio-diesel. [2M CO5 BTL1]

Part-B

4X12=48M

2. a) What is meant by de-ionized water? Explain the preparation of de-ionized water by Ion Exchange method. [7M CO1 BTL2]
- b) Explain any five factors effecting the rate of corrosion. [5M CO1 BTL2]
3. a) Write about Galvanic cells with suitable example. [5M CO2 BTL3]
- b) How is Lead-Acid storage battery constructed? Explain its functioning with cell reactions. [7M CO2 BTL2]
4. a) Explain Molecular orbital energy level diagram of O₂. [4M CO3 BTL2]
- b) Explain the preparation, properties and applications of Silicone rubber and Kevlar. [8M CO3 BTL2]
5. a) Describe Proximate analysis of coal and its significance. [7M CO4 BTL2]
- b) Calculate the Gross and Net calorific values of a coal having the following composition:
 Carbon = 85%, Hydrogen = 10%, Sulphur = 3%, Oxygen=2.1% Rest in ash.
 (Latent heat of steam is 587 cal/gram). [5M CO4 BTL3]
6. a) What are the sources of bio-diesel? Explain the concept of transesterification. [7M CO5 BTL2]
- b) How are the composite materials classified? Explain. [5M CO5 BTL2]
7. a) Briefly explain about LPG and CNG with their composition and uses. [4M CO4 BTL2]
- b) Write the mechanism of conduction in polyacetylene. Mention any 4 applications of conducting polymers. [8M CO3 BTL3]